

Smart Elevator Control System Simulation

An Energy-Efficient Approach to Vertical Transportation

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Project Inspiration

- Daily university elevator experience sparked innovation
- Ethical dilemma: Convenience vs. environmental cost
- Goal: Radical energy efficiency without compromising reliability
- Testing ground: Multi-story shopping mall

System Overview

Building Specifications

- 5 floors (-1, 0, 1, 2, 3)
- Max load: 800 kg
- 24/7 operation

Technical Platform

- Platform: GNU Octave GUI
- Single controller architecture
- Real-time simulation

Core Features

1. Smart Request Queue Management

2. Motion Sensor Door Control

3. Weight Monitoring (800kg limit)

4. Emergency Priority System

5. Energy-Saving Automation

6. Time-Based Adaptability

Four Interface Panels:

1. **Elevator Status Panel**
   Real-time monitoring

2. **Cabin Interface Panel**
   Passenger controls

3. **Control & Monitoring Panel**
   Simulation parameters

4. **Emergency Control Panel**
   Safety overrides

⚠ Emergency Systems

🔥 Fire

Auto-move to floor 0

🌊 Flood

Disable basement, move to floor 3

⚡ Earthquake

Immediate stop

Priority: Emergency > Manual Stop > Normal Operation

Energy Efficiency

Energy-Saving Mode Activation

Intelligent power management based on usage patterns

Time-Dependent Speeds

- Day (07:00-22:00): **4 sec/floor**
- Night: **6 sec/floor**

Weekend Energy Boost

- Friday-Saturday
- 12:00-20:00

+50% speed increase

QR Code Feedback System



- Real-time customer feedback collection
- Automatic maintenance alerts
- Song request integration
- Feedback statistics tracking

Technical Implementation

Platform

GNU Octave

Development

11 major revision cycles

Tools & Resources

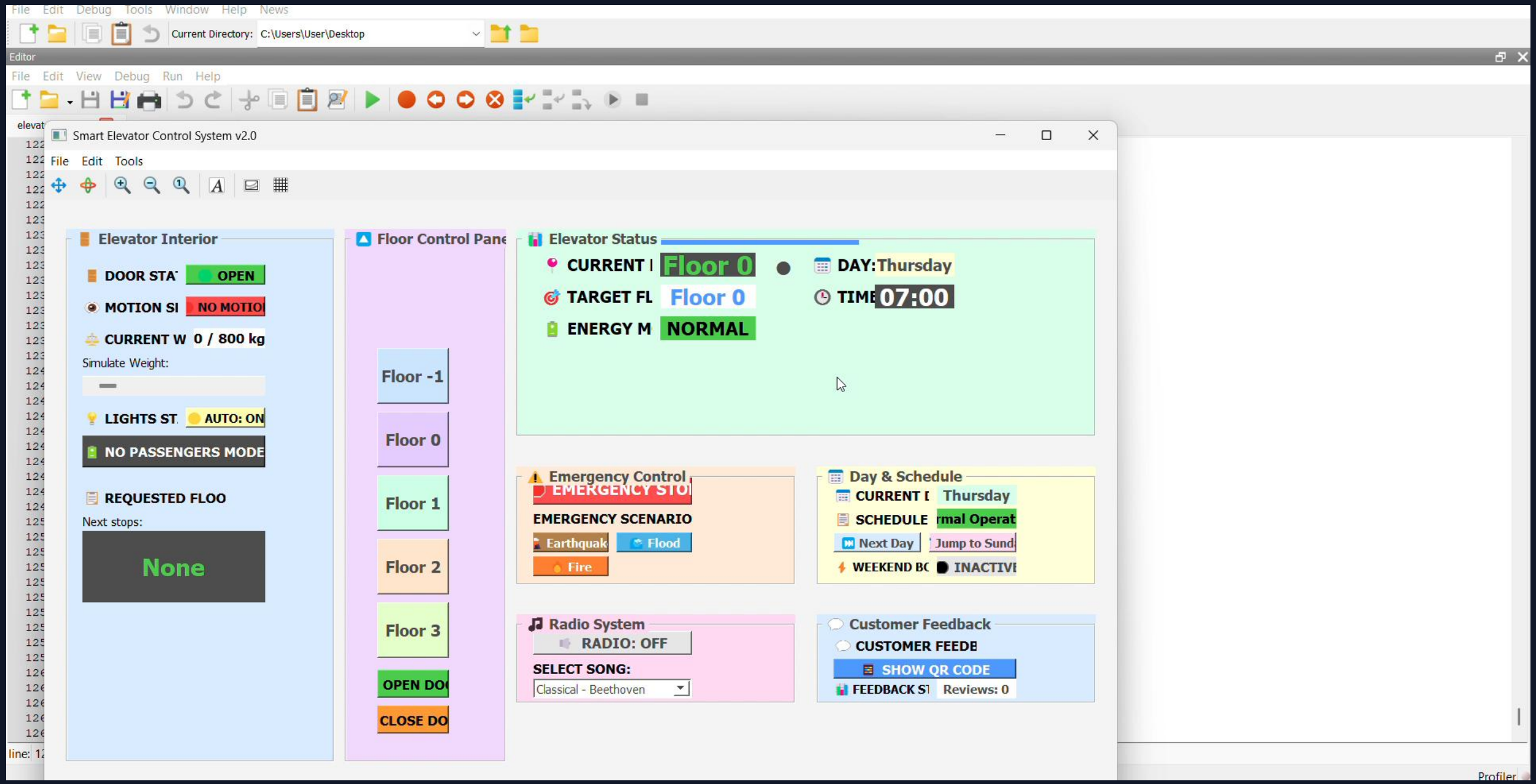
- MATLAB-style GUI
- Online Resources
- Engineering References
- Iterative Development

Educational Value

- ✓ State-based control system design
- ✓ Safety-critical decision making
- ✓ Human-machine interaction
- ✓ GUI-based simulation
- ✓ Practical engineering problem-solving

Conclusion

- Successfully demonstrated smart elevator simulation
- Balanced efficiency, safety, user experience
- Foundation for AI integration
- Contribution to sustainable design


File Edit Debug Tools Window Help News
Current Directory: C:\Users\User\Desktop
Editor
File Edit View Debug Run Help
Smart Elevator Control System v2.0
File Edit Tools
Elevator Interior
DOOR STA OPEN
MOTION SI NO MOTIO
CURRENT W 0 / 800 kg
Simulate Weight:
LIGHTS ST AUTO: ON
NO PASSENGERS MODE
REQUESTED FLOO
Next stops:
None
Floor Control Pane
Floor -1
Floor 0
Floor 1
Floor 2
Floor 3
OPEN DO
CLOSE DO
Elevator Status
CURRENT I Floor 0
TARGET FL Floor 0
ENERGY M NORMAL
DAY:Thursday
TIME 07:00
Emergency Control
EMERGENCY STO
EMERGENCY SCENARIO
Earthquak
Flood
Fire
Day & Schedule
CURRENT I Thursday
SCHEDULE rmal Operat
Next Day
Jump to Sund
WEEKEND BC INACTIVE
Radio System
RADIO: OFF
SELECT SONG:
Classical - Beethoven
Customer Feedback
CUSTOMER FEEDB
SHOW QR CODE
FEEDBACK S Reviews: 0
Profiler